

Homework: TypeScript Decision Ladders

100 points + 5 bonus points | Due August 8, 2026

Student name

Date

Part 1 - Setup Check

20 points

Node version

npm version

TypeScript version

If a command failed, record the error and your next step:

Explain the job of each tool:

Visual Studio Code

TypeScript compiler

Node.js

Blocked setup?

Use the TypeScript Playground and join the 4:30-5:30 PM Pacific Hangout Session for help. You can still earn full credit.

Workflow and Prediction

Homework continued

Part 2 - Put the Workflow in Order

20 points

- ☐ Run node decisions.js
- ☐ Save decisions.ts
- ☐ Read the terminal output
- ☐ Run tsc decisions.ts
- ☐ Type or change the TypeScript code

Complete the workflow:

decisions.ts -> -> decisions.js -> -> output

Part 3 - Predict the Branch

25 points

```
if (temperature >= 90) console.log("Very hot");
else if (temperature >= 70) console.log("Warm");
else if (temperature >= 50) console.log("Cool");
else console.log("Cold");
```

95		90		72	
70		50		49	

Why are 90, 70, 50, and 49 useful boundary tests?

Build Your Own Program

Homework continued

Part 4 - Build Your Own Program

25 points

Create homework-decisions.ts with one typed variable, one if, at least two else if branches, one final else, and a different message in every branch.

What does your program decide?

Write your final TypeScript code:

Record four tests:

Test 1		Output	
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Test 2		Output	
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Test 3		Output	
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Test 4		Output	
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AI Reflection and Submission

Homework conclusion

Part 5 - AI Reflection

10 points

An AI assistant gives you code that compiles and runs. Does that prove the logic is correct? Explain why or why not, and name two test values you would use before trusting it.

Bonus - Improve the User Experience

+5 points

Use a template literal to include the tested value in the output message.

```
console.log(`A score of ${score} earns Hero rank!`);
```

Write your improved output line:

Submit your homework to:

1. Microsoft Teams
2. Ishwari Raut ma'am

Submit the completed homework plus your .ts code or Playground screenshot.
Never include passwords or private information.